LESSON PLAN

AUXILIARY EQUIPMENT (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Efficiency, BHP, IHP, FHP, SHP, knocking and detonation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Efficiency, BHP, IHP, FHP, SHP, knocking and detonation. Ignition system, basic principles, description and operations
5. Specific Objectives	22.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine efficiency curves and ignition timing charts; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main power outputs of a piston engine?
(b) Define the term 'knocking' in the context of combustion.
(c) Why is the ignition system critical for engine operation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Brake Horse Power definition • Indicated Horse Power calculation • Friction Horse Power relationship • Shaft Horse Power output • Engine efficiency types	L & Ds	PPT, IB	15 min
(b)	• Knocking causes and effects • Detonation vs normal combustion • Ignition system basic principles • Spark plug operation description • Ignition timing adjustments	L & Ds	PPT, PDF	20 min
(c)	• Fuel octane rating impact • Cooling system role in detonation • Ignition system components overview • High tension lead function • Distributor operation basics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between IHP, FHP, and BHP.
(b) Describe the difference between knocking and detonation.
(c) List three factors that influence ignition timing.

12. Home Assignment and Follow Up	Prepare a brief note on the limitations of simple float type carburettors to be discussed in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Description & operation of simple float type carburettor, modified carburettor, limitation Types of Engine starters, & trouble shooting on piston engine	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min